Shape Activated CAM Learning for Weakly Supervised Remote Sensing Semantic Segmentation

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***Abstract*— Class activation map (CAM)-based weakly supervised semantic segmentation (WSSS) of remote sensing (RS) images has attracted extensive research interests for its potential in reducing annotation cost. However, challenged by unconstrained activation issue, existing methods struggle to delineate object boundaries clearly, making them particularly difficult to separate multiple densely packed objects, which are common in RS images. By conducting an in-depth analysis of RS image characteristics, we observed a strong correlation between object shapes and their semantics. Inspired by this finding, we propose an intrinsic shape activation network (ISANet) to learn the category-relevant shape priors as geometry constraints for target-focused region activation in WSSS of RS images. The key idea is to distill the intrinsic shape priors from the hybrid features that are deterministic in classification. Specifically, we adopt a dual-branch architecture to decouple the learning of shape and texture features and leverage a shape awareness alignment module (SAM) to generate boundary-clear CAMs for computing pseudo-labels. In this way, CAMs are generated with perception of target shapes, which increases the completeness of activation regions and alleviates the ultrarange responses. Extensive experiments demonstrate the superiority of our method in delineating densely packed objects with clear contours, which is especially beneficial for separating multiple targets in RS images. Our method improves the mean intersection over union (mIoU) of the state-of-the-art method by 7.9% and 3.3% on the NWPU VHR-10 and iSAID dataset, respectively.**

***Index Terms*— Class activation map (CAM), remote sensing (RS), shape prior, weakly supervised semantic segmentation (WSSS).**

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I. Introduction

SEMANTIC segmentation of remote sensing (RS) images aims to assign categorical labels to each pixel, facilitating a wide spectrum of applications, including urban planning, environmental monitoring, and disaster management [1], [2], [3], [4], [5], [6], [7]. In recent years, deep learning has been extensively studied with fully supervised learning models for RS image semantic segmentation [8], [9], [10], [11]. Despite the strong performance demonstrated by fully supervised methods, they depend heavily on expensive pixel-level annotations for training semantic segmentation models. The special characteristics of RS images, i.e., high spatial resolution and the wide coverage make it particularly time-consuming and labor-intensive to create high-quality pixel-level labels for them, which severely limits the application scenarios of fully supervised semantic segmentation [12], [13], [14]. To alleviate the intensive workload of manual pixel-level annotations, a growing number of researchers have recently shifted their focus toward investigating weakly supervised semantic segmentation (WSSS), which relies on low-cost image-level annotations [15], [16], [17], [18], [19].

WSSS typically involves three main steps [20], [21]. First, image-level labels are employed to supervise the training a classification network, which generates a class activation map (CAM) [22] as the seed region. The seed region is then refined and diffused by predicting semantic affinities between adjacent pixels to derive detailed pseudo-labels. These pseudo-labels are finally utilized to train a network for semantic segmentation. In WSSS, the quality of the generated CAMs in the initial step has a significant impact on the segmentation results. Due to the limited receptive field of convolutional neural networks (CNNs), coarse CAMs often focus on the most discriminative regions of a target, which severely limits the model's ability to accurately localize object boundaries. To solve this problem, many attempts have been made to gradually expand the response region by iteratively erasing the discriminative regions in the training phase, so that high-recall pseudo-labels can be obtained from the fully covered targets [23], [24], [25], [26]. However, these unconstrained expansion methods do not consider the morphological characteristics of objects such as their local boundary and global shape features, which often lead to overactivation. As depicted in Fig. 1(a) (left), the CAM

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of the bird expands to the branches beyond the boundary of the bird.

Recent studies on general WSSS with natural images propose to excavate some morphological information to constrain the expansion of CAMs [27], [28]. Specifically, these approaches first extract a small set of credible boundary pixels by filtering coarse CAMs. These reliable boundaries are then used as supervision to train an edge extraction network, which aims to capture global shape information that subsequently guides the propagation of CAMs. When dealing with an individual object, these methods can efficiently extend the local boundary to a global shape. As illustrated in Fig. 1(a) (right), the bird region in the refined CAM is completely activated under the synthesized local boundary constraint.

The general boundary-constrained expansion for CAM generation of natural images, however, is not suitable for RS images. Objects in an RS image are usually densely packed, as shown in Fig. 1 (left). In the general solution, local object boundaries are simply extracted by threshold filtering, which can result in several instances being contained by a single enclosed boundary in the feature map. As shown in Fig. 1(b) (left), due to the indistinguishable activated values between the intermediate background of adjacent storage tanks and the foreground target region, the use of simple threshold filtering fails to extract the intact boundary of each object. Instead, only the exterior boundaries of multiple dense objects are derived. In such case, it is easy to confuse adjacent objects during the process of extending local boundaries to global shapes, which leads to "glued objects," where adjacent objects are mistakenly combined into a single annotated object in the pseudo-label, as depicted in Fig. 1(b) (right).

In RS images, we observe that there exists a shape consistency prior based on the statistics of the object shapes. As depicted in Fig. 1(d), we find that natural images often capture objects from various perspectives, leading to large variations in the shape of objects. In contrast, RS images offer a bird's eye view of the Earth's surface, which minimizes the perspective distortion, manifesting a high shape consistency for objects of the same category. In other words, shape is a more inherent property of RS objects and has a strong correlation to the semantics, i.e., objects with the same semantic category tend to exhibit similar shapes. However, the locations, textures, and colors can be varied for objects even within the same category. Inspired by these findings, we are motivated to leverage shape information to guide the network classification, thereby better distinguishing instances arranged closely together. However, vanilla CNNs rely on a mixture of texture, color, and shape features for classification, which hampers the learning of consistent shape representations. As shown in Fig. 1(b), mixed feature learning leads to blurred boundaries and highly inconsistent global shapes among objects within the same class. This contradicts the expected correlation between object semantics and their shapes. Therefore, our key idea is to separate shape features from the mixed ones to allow the network to implicitly learn the intrinsic connection between shape and semantics for RS objects. As illustrated in Fig. 1(c), we can achieve independent learning of shape features under the guidance of class supervision through feature decoupling, which enhances the attention to shape information. This strategy helps in generating CAMs with clear and consistent boundaries.

Following the insights from our observation and analysis, we propose an intrinsic shape activation network (ISANet) to learn to adaptively balance the importance between shape and texture information, and thus enable the network to establish the correlation between shape and semantics for addressing the feature confusion problem for RS images WSSS. As shown in Fig. 1(c), ISANet is designed to excavate the inherent category-related shape priors in RS image features, with only image-level supervision by a deep classification network. Specifically, ISANet uses Gaussian kernels to force low-level shape clues to be separated from the hybrid features that mix context and geometry information in the classification network. The extracted multiscale shape features are then merged into the shape awareness alignment module (SAM) to improve shape recognition accuracy and avoid interference from irrelevant backgrounds. Finally, ISANet learns semantic-aware shape priors under the guidance of a standard image classification loss function. Benefiting from the high intra-class geometry consistency among the targets in RS images, the proposed ISANet can easily obtain the category-related shape priors that holistically delineate the object geometries with only image-level guidance. The learned shape priors are then used to provide geometry constraints for predicting the response regions in CAMs. In this way, CAMs are generated with a high awareness of target shapes, which increases the completeness of responding regions. With such shape-conditioned CAMs, the object masks in pseudo labels can be delineated with clear contours, which is especially favorable for separating multiple targets in RS images.

Our contributions are listed as follows.

1) We experimentally validate the intraclass shape consistency and the high correlation between shapes and semantic categories among the objects in RS images.
2) We build an ISANet to learn the category-related object shape priors in RS images with image-level supervision and leverage the learned shape priors to constrain the CAM generation.
3) Extensive experiments on two challenging RS images semantic segmentation benchmarks show that our ISANet achieves state-of-the-art performance for RS image WSSS with a simple segmentation model. Our method improves the mean intersection over union (mIoU) score by 7.9% and 3.3% on the NWPU VHR-10 and iSAID datasets, respectively.

II. Related Works

A. Weakly Supervised Semantic Segmentation

In image-level WSSS, the predominant strategy entails using the CAM obtained from the classification network as the seed region to generate pseudo-labels. The pseudo-labels are then utilized to supervise the training of a segmentation network. The core of this process is to generate high-quality CAMs. However, limited by the image-level supervision, CAMs derived from classification networks tend to merely focus on the most discriminative region of an object, leading to an incomplete activation issue.

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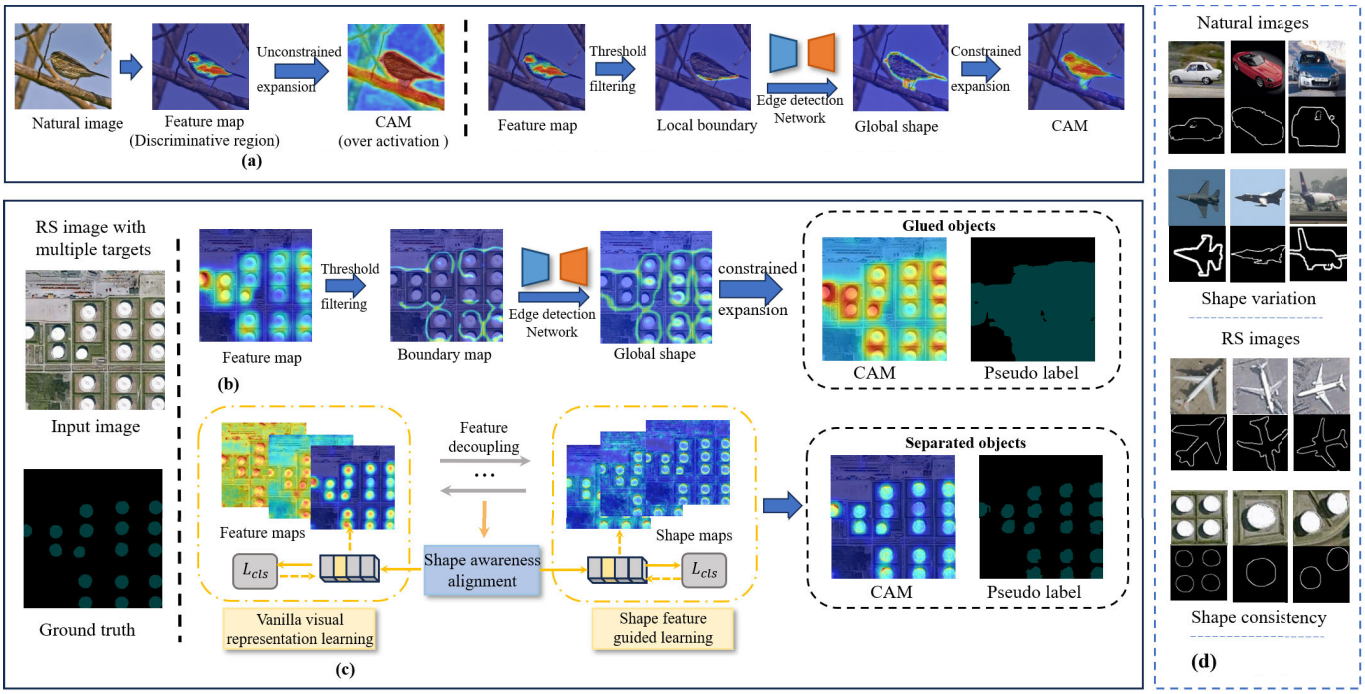


Fig. 1. Issues caused by unconstrained and boundary constrained CAM generation. (a) General WSSS based on unconstrained and boundary constrained expansion for natural images. (b) General WSSS based on boundary constrained expansion cause confusing adjacent boundaries and gluing objects in RS images. (c) Proposed ISANet is capable of clarifying adjacent boundaries and separating multiple objects in RS images. (d) Comparison of the shape consistency between natural images and RS images.

To tackle the aforementioned problem, many efforts have been made in the field of WSSS. Some earlier methods were built with erasure mechanisms to enable the expansion of responses in CAMs. Adversarial erasing network (AE) [23] introduced an adversarial erasing strategy, which iteratively masks the explored target regions to drive the classification network to discover new target areas. Adversarial Complementary Learning network (ACoL) [24] proposed an end-to-end learning framework, which utilized complementary target regions for classification and jointly produced a comprehensive target localization. Self-erasing network (SeeNet) [25] further refined the erasure mechanism by employing a self-erasing strategy to prevent activation expanding into the background. Class-specific adversarial erasing framework (CSE) [26] randomly masks an object class based on CAM and then explicitly penalizes predictions of the erased class. In this way, the activated region of an object in the CAM can better fit to the boundary of the object on the image. However, these methods are often confronted with the problems of low-efficiency and over-erasing, especially for small objects.

Since the first step of pseudo-label calculation is the filtering of responses in CAMs with simple thresholding, during which boundary clues in CAMs can be easily discarded. Therefore, many works attempt to refine CAMs to obtain complete object regions by imposing constraints on the morphological boundary. Seed, expand and constrain (SEC) [29] utilized a generation network to constrain the expanded regions to align with the object boundary during the extension of the initial seed regions from precomputed CAMs. Pixel-level semantic affinity (PSA) [30] proposed to learn pixel affinities to propagate semantic information from strongly responsive regions in CAMs to neighboring pixels. Inter-pixel relation network (IRN) [31] developed an inter pixel relation learning network to calculate class boundary maps based on pixel affinities. Meanwhile, boundary exploration based segmentation (BES) [32] leveraged the boundary clues in the original CAM as auxiliary supervision at the segmentation stage, and boundary-enhanced co-training (BECO) [28] introduced a co-training approach that involves two interactive networks to enhance uncertain pixel learning and developed a boundary-enhancement tactic leveraging dependable forecasts to create synthetic boundaries for challenging edge regions. Subsequently, online attention accumulation network (OAANet) [33] accumulated attention maps from multiple training stages to the obtained integral object regions. Explicit pseudo-pixel supervision (EPS) [34] used the preexisting saliency maps to supervise region learning to produce high-quality attention maps.

However, the morphological constraint-based methods only consider the local boundary information but neglect the importance of global shape, which leads to object gluing in the RS images. Considering that objects in RS image have high intraclass shape consistency, we believe that it is possible to excavate this prior for implicitly learning the global shape of RS objects within each semantic category. In this way, the expansion for the activated object regions in CAM can be constrained by the global shape of each object, during which the RS objects in a CAM are segmented with a clear boundary and high completeness.

B. Weakly Supervised Semantic Segmentation in Remote Sensing Images

With the rapid development of WSSS for natural images, increasing interests have now been paid to the study of WSSS for RS images. Weakly supervised feature-fusion network (WSE-NET) [15] and superpixel pooling and multi-scale

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feature-fusion network (SPMF-Net) [16] were two pioneering works that focus on RS binary segmentation. WSF-NET devised a feature fusion network to represent the target features of RS images and implemented a balanced binary training strategy to rectify the class imbalance. SPMF-Net employed a super-pixel pooling operation and a multiscale feature fusion module for building segmentation. To address multicategory WSSS problem in RS images, siamese network based on the explicit pixel-level constraints (SEPC) [18] proposed a siamese network with an explicit pixel-level constraint framework to enhance the segmentation accuracy of multiclass objects. Since previous methods only consider single-image information, cross-image semantic mining framework (CISM) [17] introduced a cross-image semantic mining framework to facilitate the generation of more accurate and fine-grained CAMs for WSSS. Pseudo label noise suppression (PLNS) [35] leverages a self-supervised encoder fine-tuned on RS datasets to provide additional feature similarity information, thereby reducing the noise in the pseudo-labels.

Although achievements have been made by existing methods in WSSS for RS images, they still leave a challenging issue of unconstrained response expansion in CAM, which can incur object gluing in pseudo-labels for RS images containing multiple densely packed objects. In this study, we observed that there exists a high correlation between the shape and semantic category of RS objects. Based on such correlation, we designed ISANet to incorporate the shape consistency prior of RS objects from realizing constrained response computation. In this way, CAMs can be generated with clear object boundaries for producing high-quality pseudo-labels.

III. Methodology

In this section, we provide a detailed introduction of the proposed ISANet. We start by outlining the preliminary of CAM generation. Subsequently, we elaborate on the insight and motivation behind the design of ISANet. Finally, we provide a detailed description the framework of ISANet and its core components.

A. Preliminary

CAM is an attention map that highlights which regions of an image are key for network decisions, e.g., for classification. It is widely used to obtain coarse seed regions in WSSS. Given an input image x and its category label $y \in \{0, 1\}^N$, we denote the feature map extracted from the final convolutional layer of a trained network as $f(x) \in R^{h\times w\times d}$. The training of the network is guided by binary crossentropy loss (BCE), which is formulated as follows:

$$\mathscr{L}_{\text{bce}} = \frac{1}{K}\sum_{k=1}^{K} y_k \log \sigma(z_k) + (1 - y_k)\log[1 - \sigma(z_k)] \quad (1)$$

where z_k denotes the prediction logit of the kth class, $\sigma(\cdot)$ is the sigmoid function, and K is the total number of foreground object categories. $y_k \in \{0, 1\}$ is the image-level label for the kth class, where 1 denotes that the class is presented in the image. For a certain class k, CAM can be calculated as follows:

$$\text{CAM}(x)_k = \frac{\text{ReLU}(M_k)}{\max(\text{ReLU}(M_k))}, \quad M_k = w_k^T f(x) \quad (2)$$

where w_k denotes the weights of the linear classifier for class k, and rectified linear unit (ReLU) and max functions are used to filter out negative values and normalize CAM values into $[0, 1]$.

B. Insight and Motivation

In WSSS, it is vital to obtain the precise and complete activation of object regions in CAMs. This requires that the network learn to accurately localize the target objects and correctly identify their semantics with only image-level supervision. The shape of objects inherently imposes constraints on their precise localization. Hence, if the network can interpret the semantic categories of objects through shape information, it would be promising to leverage shape clues to activate object regions with correct semantics and clear boundaries. However, in natural images, the shapes of objects vary significantly, even among objects within the same semantic category, making them impractical to be used as a basis for recognition. Fortunately, from the examples presented in Fig. 1(d) in Section I, we found that objects with the same semantic category tend to exhibit similar shapes in RS images. This suggests that we can exploit the consistency of RS objects and their high correlation with semantic categories to fully activate objects in CAMs. In the following, we provide a more comprehensive qualitative and quantitative analysis to validate the existence of shape consistency and demonstrate its feasibility for inferring the semantics of RS objects.

For qualitative analysis, Fig. 2 displays samples randomly selected from natural and RS image datasets. There exists a pronounced disparity in the morphological characteristics of targets between natural images and RS images. In natural images, the shapes of objects, such as aircraft and vehicles, are inconstant. Conversely, in RS images, the shapes of corresponding objects exhibit a remarkable consistency and uniformity. This observation suggests that shape is a more inherent property of RS objects and has stronger correlation to the semantics.

To verify the existence of shape consistency and the association between shape and category more accurately and comprehensively, we provide a statistical analysis on the shapes of RS objects. We first need to quantify the similarity of shapes among instances in the same category. Instead of calculating the shape similarity between each pair of instances, we take an indirect approach by utilizing a circle as the shape reference and construct a shape descriptor that measures the difference between the shape of each instances and the reference circle. In detail, we employ the reciprocal form of the iso-perimetric quotient (RIPQ) [36] measurement as the shape descriptor, which is defined by $\text{RIPQ} = (P^2/4\pi A)$, where P and A are the perimeter and area of the shape, respectively. According to the isoperimetric inequality $4\pi A \leq P^2$, as the shape becomes more circular, the descriptor approaches a value of 1. Thus, this descriptor can be used as an effective indicator to measure the similarity between different shapes.

We provide the statistics of the shape descriptor values (RIPQ) of instances from the natural image dataset VOC2012 [37] and two RS image datasets, NWPU VHR-10 [38] and iSAID [39] in Table I and Fig. 3. Plane and vehicle are the two categories shared by both datasets. In Table I,

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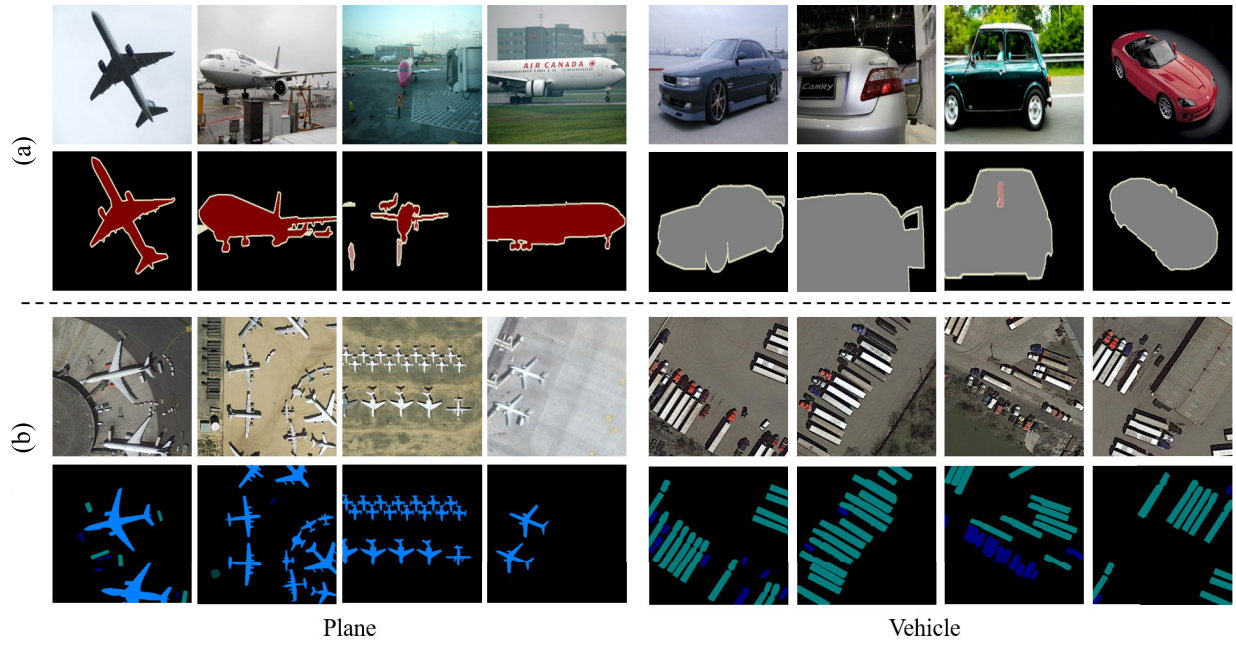


Fig. 2. Comparison of the geometry consistency between natural images and RS images. We visualize original image (odd rows) and the corresponding semantic annotation (even rows) on the natural dataset and RS dataset. (a) Shape consistency of objects in natural images is heavily destroyed by view diversity and perspective distortion. (b) Objects in RS images share high intraclass geometry consistency due to the special data collection viewpoint.

TABLE I

STATISTICAL ANALYSIS OF THE VARIANCES OF BOTH CATEGORIES OF RIPQ, AND THE AVERAGE VARIANCES OF ALL CATEGORIES

Dataset	Vehicle	Plane	Mean var
VOC12	1.77	0.96	2.15
NWPU	0.01	0.83	0.20
iSAID	0.03	0.67	0.28

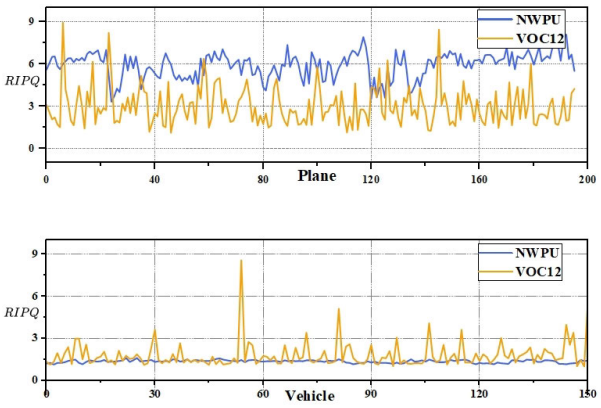


Fig. 3. RIPQ metric of randomly selected planes and vehicles. Regardless of the category, the blue RS images exhibit significantly less fluctuation compared to the yellow natural images.

we calculate the variance of RIPQ of each category and the mean variance of all categories in each dataset. According to Table I, objects in RS images demonstrate a much higher geometry consistency. We also plot the RIPQ curves of the two shared categories in Fig. 3, which perceptually demonstrates the higher geometry consistency among the instances in RS images with a smoother RIPQ curve.

Based on the statistical analysis conducted above, we have validated the strong correlation between object shapes and their semantics in RS imagery. Now, we aim to explore whether the shape features can serve as the principal cues for inferring the semantics of RS objects, thus alleviating the unconstrained activation issue caused by the mixed learning of texture, color, and shape features in WSSS. To achieve this, we further conducted experiments on the three previously used datasets, i.e., the NWPU VHR-10, iSAID, and VOC12, with a ResNet50-based [40] image classifier that is pretrained on ImageNet 1k [41]. Specifically, we first performed a standard training on the three datasets with the RGB images as input. Additionally, we trained the image classifier with only the binary shape maps as input. The shape maps contain no texture or color information, but only object edges extracted from the pixel-wise semantic segmentation labels. For a fair comparison, the shape maps were duplicated into three-channel images to match the input structure of RGB images. All models were trained using a stochastic gradient descent (SGD) optimizer with multilabel soft margin loss. The classification performance of all models was evaluated with the average precision (AP) metric. The numerical results are shown in Table II.

TABLE II

AVERAGE ACCURACY SCORES ON VOC12 AND TWO RS DATASETS: NWPU VHR 10 AND iSAID

Dataset	Plane	Vehicle	mAP
NWPU RGB	100	100	97.4
NWPU edge	100(-**0**)	94.5(-**5.5**)	90.6(-**6.8**)
iSAID RGB	97.65	85.72	85.36
iSAID edge	95.81(-**1.84**)	81.82(-**3.9**)	76.10(-**9.26**)
VOC12 RGB	97.8	89.0	88.7
VOC12 edge	84.5(-**13.3**)	62.0(-**27.0**)	59.0(-**29.7**)

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As shown in Table II, the model trained using only shape maps achieved performance nearly identical to the original model trained on RGB images for the NWPU VHR-10 dataset and the iSAID dataset (RS dataset). However, for the VOC12 dataset (natural dataset), the results differed significantly. The model trained with only shape maps shows a notable performance drop, with a decrease of nearly 30% in the overall mean AP (mAP) compared to the model trained with RGB images. These results suggest that the classifier can largely infer semantics from shape features alone for RS objects, which exhibit good shape consistency. In contrast, it struggles to preserve this capability for natural objects, which demonstrate drastic shape variations.

Grounded in the above insights and preliminary experiments, we are motivated to leverage shape priors for more accurate and complete activation of RS objects in CAMs to enhance the robustness of WSSS. In the previous semantic segmentation experiments, shape maps were directly derived from the pixel-wise supervision labels. However, in WSSS, only image-level labels are available. To address this challenge, we propose the ISANet, which effectively separates the shape features from the mixed features and thereby learns the strong correlation between object shapes and their semantic categories. The details of ISANet are discussed in Section III-C.

C. Intrinsic Shape Activation Network

In this section, we will introduce the proposed ISANet and its core components. The architecture of ISANet, illustrated in Fig. 4(a), follows a dual-branch design with N paired CNN and shape stages. The number of stages N is adjustable based on the selected backbone. Given an input RGB RS image, the CNN branch is designed to extract semantic regional features with coarse object localization, while the shape branch focuses on distilling boundary-related features to refine object localization. In RS images, the presence of numerous non-target objects and background noise significantly interferes with the learning of both the semantic and boundary features for target objects, as image-level labels cannot provide the fine-grained semantic and boundary supervisory signals required for accurate learning. To address this, the two branches interactively gather and exchange informative features during training, enhancing WSSS-specific learning under image-level supervision, as illustrated in Fig. 4(b). A SAM is appended on top of the final pair of CNN and shape stage to provide pixel-level constraints for the classifier, which bridges the gap between the image-level supervision and pixel-level predictions, as shown in Fig. 4(c). In the following, we will describe each module of our framework in detail.

Fig. 4(b) illustrates the dual branch structure of each pair of CNN and shape stage. The CNN branch mainly consists of multiple stacked ResNet blocks, which can be replaced with any backbone network, e.g., VGG16 and EfficientNetV2. Here, we choose ResNet50 as the backbone network, considering its wide range of uses in the task of WSSS. The shape branch is mainly composed of a shape extractor with some designed convolutional operations. The two branches are connected by two 1×1 convolution layers with up and downsampling operations to strengthen the learning of shape features and the shape-constrained regional features in each stage.

Specifically, the CNN branch first enhances the input features from the previous stage using a 1×1 convolution layer. For convenience, we denote the output features as $m_i \in R^{C \times H_i \times W_i}$, where i is the stage index of the CNN branch. The features m_i are then processed by several ResNet blocks to extract high-level semantic region features. Simultaneously, features m_i are passed through another 1×1 convolution layer, followed by an upsampling operation. While the CNN branch extracts general high-level visual features, convolutional downsampling can blur object edges. The upsampling operation helps preserve more edge details in the high-resolution feature map and enables the shape branch to maintain a consistent spatial resolution of 128×128 pixels throughout the processing pipeline. The upsampled features are then fused with the shape features by element-wise addition to suppress non-target region responses while emphasizing shape-related region responses. The refined features are processed by another 1×1 convolution and upsampling operation before being fused with the general semantic region features derived from the ResNet stage, thus generating more representative shape-related region features for the next stage.

In the shape branch, as shown in Fig. 4(b), the shape features from the previous stage are first fused with the upsampled features from the CNN branch. Notably, in the first stage of the shape branch, we directly use the features obtained from the input image after it passes through the first convolutional layer, the batch normalization layer, the ReLU activation, and the max pooling layer of ResNet. Compared to the shape features from the previous stage, the fused features focus more on the prominent semantic regions. We denote the fused features as α_i. To disentangle shape information from the fused features α_i, we process it by a shape extractor, as shown in Fig. 5, to generate the shape mask b_i, which is formulated as follows:

$$b_i = \|\alpha_i - \alpha_i * w\| \tag{3}$$

where $*$ denotes the convolution operator and w denotes the Gaussian kernel. The use of the subtraction operation helps preserve the response of the boundary regions of objects, while smoothing the inner regions of objects and reducing the noisy areas in the obtained shape mask. With the shape mask b_i, shape information is ready to be embedded into ith shape feature $\hat{\alpha}_i$, which is computed by

$$\hat{\alpha}_i = b_i \odot \alpha_i \tag{4}$$

where $\odot$ means element-wise multiplication. All of the processes described above constitute one shape stage. A total of N shape stages form the shape branch, which leverages the extracted shape information to learn shape priors and assign categories to these shapes, guided by category labels. Due to the strong intraclass shape consistency, as demonstrated in Section III-B, the shape branch can effectively converge and accurately locate target shapes in RS images using image-level category supervision.

To obtain more refined shape features, our ISANet employs a higher resolution in the shape branch. Therefore, upsampling and downsampling modules are used during the interaction

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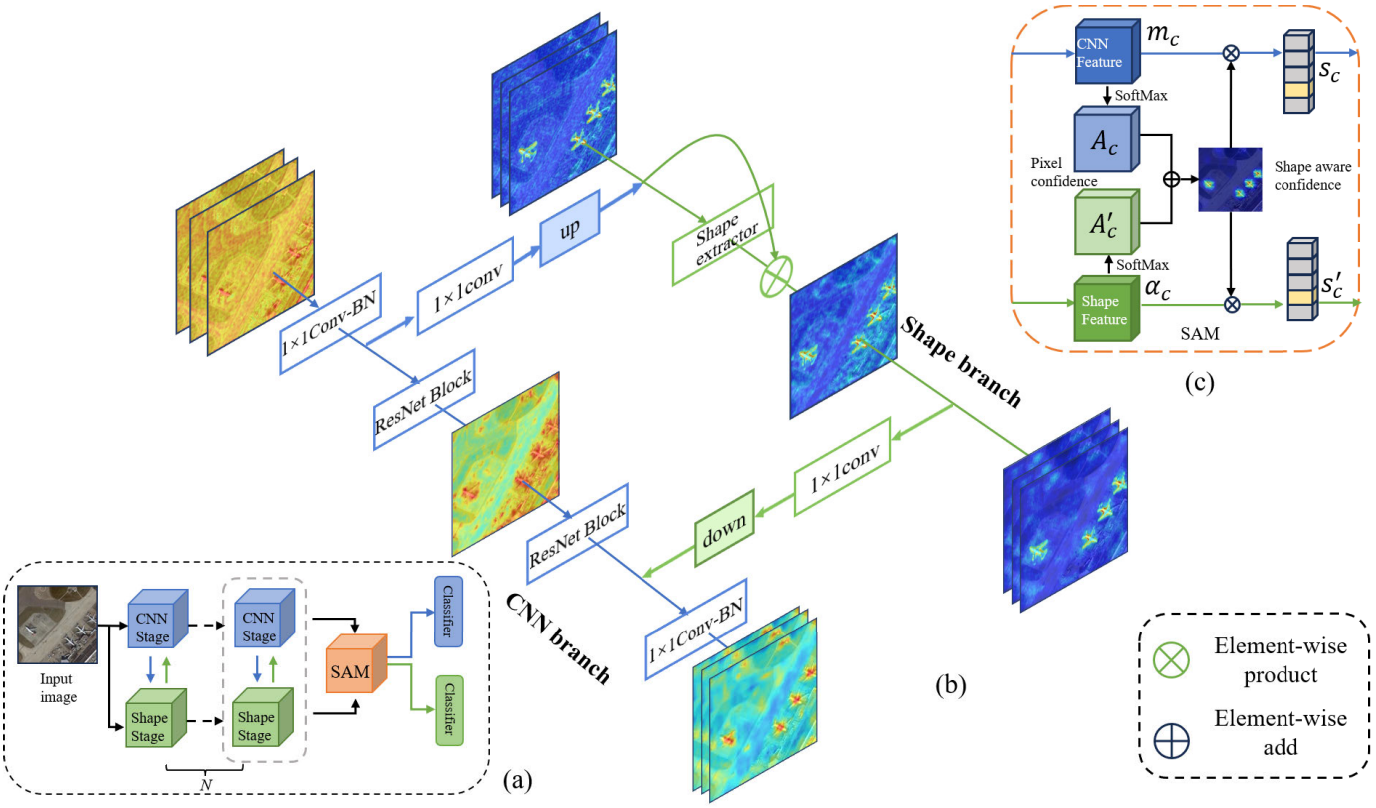


Fig. 4. Network architecture of the proposed ISANet. (a) Framework of ISANet. (b) Internal structure of a paired CNN and shape stage. The feature maps of the CNN stage are fed into a specially designed shape block, which separates shape cues from the hybrid features. (c) Structure of the SAM. The GAP layer is replaced with the weighted average pooling, and the weights are determined by the shape feature and the classical feature.

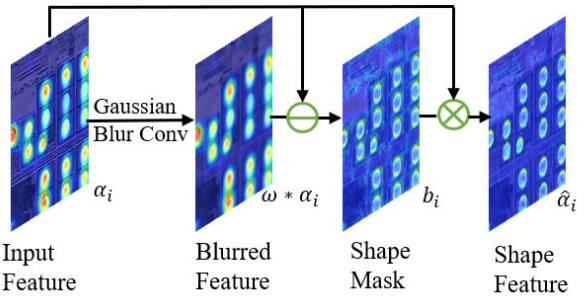


Fig. 5. Illustration of shape extractor. This module separates input features using Gaussian convolution, where $\ominus$ and $\otimes$ represent element-wise subtraction and element-wise product, respectively.

between the two branches to resolve any misalignment. During this interaction, the classical branch, which excels in developing a high-level semantic understanding of the scene, guides the shape branch to focus on the relevant targets in the image. Conversely, the shape branch encourages the classical branch to extend its focus beyond the most discriminative parts of the target. Through this interaction, both branches synergistically activate complete object regions, providing refined features for the subsequent SAM.

As shown in Fig. 4(c), the SAM deploys a new attentional pooling structure to enhance both the pixel-level constraint capability of the network and the synergy between the two branches. We remove the global average pooling (GAP) layer for obtaining classification scores in ResNet and use the attention pooling instead. This approach allows dynamic assignment of different weights to individual pixels. Considering the complementarity between the regions focused on by the CNN branch and the shape branch, we combine the weights calculated by both branches and assign them to each pixel. Let m_c and α_c represent the feature maps of the CNN branch and the shape branch for channel c, respectively. The corresponding pixel confidence maps A_c and A'_c are generated by using the SoftMax function on the feature maps m_c and α_c

$$A_{c,i} = \frac{e^{km_{c,i}}}{\sum_j e^{km_{c,j}}}, \quad A'_{c,i} = \frac{e^{k\alpha_{c,i}}}{\sum_j e^{k\alpha_{c,j}}} \tag{5}$$

where k denotes a hyperparameter to adjust the intensity of the SoftMax function. When k is set to 0, the confidence map assigns equal weights to each pixel. As k increases, pixels with high response values derive higher confidence values.

The value at each position in the confidence maps indicates the contribution of the corresponding position in the feature maps to the classification scores s_c and s'_c, which are calculated by the weighted sums of the pixel confidences from both branches, as follows:

$$s_c = \sum_i m_{c,i} \frac{A_{c,i} + A'_{c,i}}{2}, \quad s'_c = \sum_i \alpha_{c,i} \frac{A_{c,i} + A'_{c,i}}{2}. \tag{6}$$

SAM provides pixel-level constraints for the classifier to overcome the limitation of image-level prediction and highlights the contribution of the two branch activated regions to the final classification score. Through this interaction process, the

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two branches synergistically activate the complete target area, providing more information for the generation of CAMs.

Subsequently, with the hybrid features m_n from the classical branch and geometry clues $\hat{\alpha_n}$ from the shape counterpart, the CAM is obtained by

$$\mathrm{CAM} = \frac{\mathrm{ReLU}\left(w_c^T m_n\right)}{\max\left(\mathrm{ReLU}\left(w_c^T m_n\right)\right)} + \frac{\mathrm{ReLU}\left(w_s^T \hat{\alpha_n}\right)}{\max\left(\mathrm{ReLU}\left(w_s^T \hat{\alpha_n}\right)\right)} \quad (7)$$

where w_c^T and w_s^T denote the classical and shape branch classifier weights, respectively.

IV. Experiments

To validate the effectiveness of the proposed ISANet for WSSS, we evaluate its performance on two widely used RS semantic segmentation datasets: NWPU VHR-10 [38], iSAID [39], and ISPRS Vaihingen datasets [42]. In the following, we will first give a detailed introduction of the two datasets and the implementation settings. We will then present a comprehensive comparison with state-of-the-art WSSS methods. Finally, we give an extensive analysis on the design of the proposed ISANet by ablation studies.

A. Dataset and Evaluation Metrics

1) NWPU VHR-10: NWPU VHR-10 dataset is originally an RS image object detection dataset, which has been extended for WSSS with the segmentation annotations provided by Su et al. [43]. The dataset contains 715 optical RS images from Google Earth with a spatial resolution of 0.5–2 m and 85 pan-sharpened color infrared images from the Vaihingen dataset with a spatial resolution of 0.08 m. The NWPU VHR-10 dataset includes ten classes: The airplane (PL), ship (SH), storage tank (ST), baseball diamond (BD), tennis court (TC), basketball court (BC), ground track field (GTF), harbor (HA), bridge (BR), and vehicle (VE). Following the previous works [18], the full-resolution images are cropped into 512 × 512 with a stride of 256. The training set and the testing set contain 2916 and 728 slices, respectively.

2) iSAID: iSAID benchmark provides 1411 high-resolution RS images for training and 458 images for testing. These images are collected from Google Earth, which were mainly taken by the JL-1 and GF-2 satellites. The dataset meticulously categorizes various structures and objects, including plane (PL), baseball diamond (BD), ship (SH), ground-track field (GTF), basketball court (BC), soccer ball field (SBF), roundabout (RA), storage tank (ST), bridge (BR), large vehicle (LV), tennis court (TC), helicopter (HC), harbor (HA), small vehicle (SV), and swimming pool (SP). In accordance with the settings of NWPU VHR-10, we randomly cropped each image in the iSAID dataset into six or eight 512 × 512 patches, obtaining 10 870 slices for training and 4133 for testing.

3) ISPRS Vaihingen: ISPRS Vaihingen dataset is a benchmark dataset for semantic segmentation in RS, primarily focusing on urban and rural scenarios. It consists of 33 high-resolution aerial images captured over the village of Vaihingen, Germany, using airborne sensors. The annotations cover five semantic classes: impervious surfaces, buildings, low vegetation, trees, and cars. The original images (averaging 2500 × 2000 pixels) are typically cropped into smaller patches of 512 × 512 pixels to facilitate deep learning processing. The dataset is officially divided into 16 training images and 17 testing images, which are commonly processed to generate approximately 1106 training patches and 1268 validation patches after splitting.

4) Evaluation Metrics: For quantitative assessment of the proposed ISANet and the comparison methods on different datasets, we employ the mIoU scores and overall accuracy (OA) indicators as the evaluation metrics.

Formally, the mIoU and OA are defined as follows:

$$\mathrm{mIoU} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{IoU} = \frac{1}{N}\sum_{i=1}^{N} \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \quad (8)$$

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}+\mathrm{TN}} \quad (9)$$

where N is the number of categories and TP indicates the count of positive samples correctly identified as positive, while FP refers to the count of negative samples inaccurately classified as positive. Conversely, TN represents the number of negative samples correctly identified, and FN signifies the count of positive samples that are incorrectly labeled as negative. Through these indicators, we offer a comprehensive evaluation of our model's performance.

B. Implementation Details

The proposed ISANet is implemented with Pytorch framework on a single NVIDIA GTX 3090 GPU. Following previous works [31], [44], we utilize a pretrained ResNet-50 model as the backbone of our architecture. We set the size of the Gaussian kernel in the shape branch to 5 and the hyperparameter k in the SAM to 10^{-3}. We only use the image-level label to train our ISANet with SGD optimizer that has a momentum of 0.9 and a weight decay of 5×10^{-4} for 20 epochs on both iSAID and NWPU VHR-10 datasets. The learning rates are initiated to 0.05 and 0.01 for iSAID and NWPU VHR-10, respectively. We employ the polynomial decay strategy [45] for learning rate adjustment to refine the learning process gradually.

For the generation of pseudo-labels, we employ multiscale testing and generate coarse pseudo-labels through threshold-based filtering on the CAM. Following prior WSSS studies [31], [46], we apply dense conditional random field (denseCRF) to post-process these coarse pseudo-labels, producing refined pseudo-labels.

In the second phase of WSSS, we simply use DeepLab V2 [20] as the segmentation model for all comparison methods. We train it for 40 000 iterations on both datasets, employing a learning rate of $2.5e^{-3}$ and a batch size of 6. The optimization process is guided by pixel-level pseudo-labels generated during the initial phase of our framework.

C. Results on NWPU VHR-10 Dataset

To verify the effectiveness of the proposed ISANet, we first conducted experiments on the NWPU VHR-10 [38] dataset. The state-of-the-art WSSS models, including IRN [31] and

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local prototype CAM (LPCAM) [44], were chosen for comparison. As these models were originally developed for natural datasets, we retrained them on the NWPU VHR-10 dataset to ensure a fair comparison. For a comprehensive comparison, the performance of all models was evaluated from three perspectives: the CAM generation, the pseudo-label generation, and the final WSSS. The quantitative results are listed in Tables III–V.

As shown in Table III, ISANet significantly outperforms all comparison methods in terms of CAM generation. Specifically, ISANet achieves a substantial gain in mIoU, ranging from 34.21% to 7.89% over existing methods. Notably, ISANet delivers the best performance across nearly all categories except for GTF. Particularly for objects with distinctive shapes, such as baseball diamonds (BD), bridges (BR), and storage tanks (ST), ISANet demonstrates a significant improvement in IoU scores, with gains of at least 19%. This improvement can be attributed to superior ability of ISANet to capture shape cues, enabling more complete object region activation compared to existing models. As revealed in the introduction, a strong correlation exists between the shape and the semantic category of RS objects. The mixed feature learning (texture, color, and shape) employed by current deep models hampers their ability to effectively capture the shape features critical for recognizing and localizing RS objects under image-level supervision. In contrast, ISANet is specifically designed to disentangle shape features from other mixed features during training, allowing it to learn the connection between shape and semantic category more effectively. As a result, with only image-level supervision, ISANet can recognize and localize RS objects with consistent shape representations, producing high-quality CAMs. Notably, our method demonstrates competitive performance, achieving the highest mIoU of 48.79% while maintaining a reasonable training time of 16 min and 5 s. In terms of inference speed, our model performs at 22.63 frames/s, which is comparable to the IRN and significantly faster than some of the other methods.

As shown in Table IV, our method achieves a dominant 52.09% mIoU, surpassing the IRN (43.02% mIoU) by 9.07%, with breakthrough improvements in multiple key categories. Existing approaches (e.g., BES, LPCAM) suffer from incomplete activation or blurred boundaries, particularly for densely distributed objects like ship and storage tank (e.g., LPCAM yields only 2.59% IoU for ship). In contrast, our method mitigates target adhesion through shape-aware feature disentanglement, achieving 57.83% and 43.72% IoU for ship and storage tank, respectively, which represent improvements of 29.76% and 23.98% over the second method. Furthermore, our method excels in geometrically regular targets such as Baseball Diamond, attaining 78.5% IoU, highlighting its effective utilization of geometric priors. Overall, the results validate that our method, by explicitly learning shape priors of RS objects, generates pseudo-labels with sharper boundaries and more complete coverage, particularly advantageous for dense and small-scale targets.

In Table V, the overall performance of WSSS methods closely mirrors that of CAM generation. ISANet achieves the highest OA and mIoU, surpassing other methods by 1.14% and 7.93%, respectively. In the category-wise comparison, ISANet consistently outperforms most methods, particularly for storage tanks (ST), with an improvement of at least 28.86%. Notably, during the semantic segmentation phase, we employed the same DeepLabV2 model across all methods. The key difference lies in the training labels used, which are pseudo-labels derived from the CAMs generated in the first phase of WSSS. As a result in Table V, the quality of the CAMs has a significant impact on subsequent segmentation performance. With the high-quality CAM generated in the first phase, ISANet achieves more robust segmentation results compared to other methods, especially for segmentation-challenging small objects, such as ships (SH) and storage tanks (ST), where ISANet delivers promising performance. These objects are usually densely packed and tend to be glued together in the CAMs generated by existing methods due to the issue of unconstrained activation. The numerical results indicate that this issue can be effectively mitigated by isolating and independently learning shape features. We also observe that our weakly supervised method achieves 76% of fully supervised performance (mIoU 53.95% versus 71.07%) with only image-level label, demonstrating competitive efficiency. In general, the significant improvement in the WSSS task further validates that our shape-activated CAM learning method can effectively leverage the distinctive characteristics of RS images, enabling the activation of objects with complete regions and clear boundaries. Fig. 6 provides a qualitative comparison results for further visual inspection.

As shown in Fig. 6, we selected four diverse scenes with different types of objects for comparison. The first row of Fig. 6 shows that all methods are capable of locating the baseball diamond in simple scenarios. However, our method demonstrates an advantage in capturing the fine geometric structure of the baseball diamond and accurately depicts its morphological details, achieving superior segmentation performance. In contrast, as shown in the second and third rows of Fig. 6, most existing methods fail to annotate densely packed objects, often causing multiple objects to be merged together. This occurs because the CAMs generated by these methods confuse the connection areas between objects. Consequently, when pseudo-labels derived from these CAMs are used to train segmentation networks, it becomes difficult to distinguish the boundaries between adjacent objects. Benefiting from the proposed shape branch, the extracted CAM can clearly identify and separate densely distributed objects by specifically learning shape features in the shape branch. As a result, networks supervised by the pseudo-labels generated from these CAMs naturally inherit the ability to differentiate between multiple densely distributed objects, thereby improving the segmentation accuracy. In the case of isolated bridges, other methods often exhibit significant over-activation, likely because the network erroneously expands the activation region to the river areas adjacent to the bridge during CAM learning. In contrast, under the constraint of shape prior, our method effectively avoids incorrect regional associations, thereby achieving precise object region localization.

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TABLE III
COMPARISON OF CAM RESULTS ON NWPU VHR 10 WITH DIFFERENT METHODS

Method	BG	PL	SH	ST	BD	TC	BC	GTF	HA	BR	VE	mIoU	Training Time	FPS
RIB [47]	45.42	4.55	2.54	10.15	11.43	17.07	13.48	26.25	22.39	4.90	2.14	14.58	32min18s	4.12
ReCAM [48]	73.90	11.65	8.95	16.46	31.91	43.33	38.08	56.70	44.50	6.13	2.88	30.41	19min40s	20.53
BES [32]	80.47	11.52	8.90	18.42	30.99	38.32	43.24	70.49	45.57	12.10	3.65	33.06	22min09s	13.28
LPCAM [44]	79.78	9.65	5.16	21.01	28.49	40.31	44.09	72.48	48.51	12.71	3.67	33.26	18min57s	3.62
IRN [31]	88.35	17.99	14.79	23.22	45.87	49.81	57.36	**73.67**	53.23	20.14	5.52	40.90	**15min49s**	**23.17**
ISANet (ours)	**94.46**	**28.95**	**24.11**	**42.52**	**67.59**	**62.32**	**69.73**	34.08	**54.32**	**45.76**	**12.85**	**48.79**	16min05s	22.63

TABLE IV
COMPARISON OF PSEUDO-LABEL RESULTS ON NWPU VHR 10 WITH DIFFERENT WSSS METHODS

Method	BG	PL	SH	ST	BD	TC	BC	GTF	HA	BR	VE	mIoU
RIB [47]	64.01	5.58	1.21	13.84	33.87	43.85	42.18	50.53	27.26	4.56	2.28	26.29
ReCAM [48]	68.32	5.24	2.3	15.24	37	**48.72**	51.76	65.92	34.45	4.59	2.19	30.52
LPCAM [44]	73.69	5.25	2.59	17.21	20.52	45.68	56.14	79.03	39.26	10.67	3.01	32.09
BES [32]	80.08	5.91	14.56	15.36	68.88	45.1	54.35	85.23	38.86	7.52	2.82	38.06
IRN [31]	89.79	18.2	28.07	19.74	65.36	46.45	**56.89**	**86.05**	44.2	14.35	4.11	43.02
ISANet (ours)	**93.8**	**33.98**	**57.83**	**43.72**	**78.5**	48.36	53.06	56.14	**65.38**	**36.67**	**5.53**	**52.09**

TABLE V
COMPARISON OF SEGMENTATION RESULTS ON NWPU VHR 10 WITH OTHER WSSS METHODS

Method	BG	PL	SH	ST	BD	TC	BC	GTF	HA	BR	VE	OA	mIoU
Fully Supervised Semantic Segmentation:													
DeepLabV2 [20]	98.57	47.83	54.27	80.47	84.15	68.93	73.68	91.74	71.11	59.43	51.63	98.64	71.07
Weakly Supervised Semantic Segmentation:													
RIB [47]	67.48	6.99	0.62	12.54	64.69	**54.01**	67.18	70.48	31.96	4.63	2.88	69.20	34.86
LPCAM [44]	76.16	5.28	2.48	2.48	25.61	25.61	**68.63**	84.19	43.27	14.06	3.74	77.42	35.01
ReCAM [48]	70.60	6.67	2.17	11.87	62.71	49.38	64.10	74.96	37.05	4.73	2.73	55.74	35.18
BES [32]	82.77	7.63	16.42	11.97	74.90	49.44	62.30	62.30	41.44	9.76	3.70	83.66	40.66
BECO [28]	82.58	17.28	30.16	10.50	69.33	42.86	59.53	**91.78**	36.62	8.45	**4.50**	83.58	41.24
IRN [31]	92.22	30.43	28.92	15.56	70.90	46.65	64.88	85.19	45.17	22.75	3.60	92.57	46.02
ISANet (ours)	**93.42**	**37.27**	**57.86**	**44.42**	**82.44**	47.51	57.11	59.27	**62.84**	**46.67**	3.91	**93.71**	**53.95**

D. Results on iSAID Dataset

For a comprehensive evaluation, we conducted additional experiments on the challenging iSAID dataset. Unlike NWPU VHR-10 dataset, the iSAID dataset contains larger-scale RS imagery and a wider variety of target categories. We also evaluated the performance of CAM and WSSS on the iSAID dataset.

The quantitative comparison result on the iSAID dataset is presented in Table VI, where the overall performance trend is similar to that observed on the NWPU VHR-10 dataset. For CAM performance, ISANet achieves the highest prediction scores in terms of the mIoU and the majority of the category-specific IoU, as shown in Table VI. A particularly representative category is the helicopter (HC), which has a small and fine geometric structure and is absent from the NWPU VHR 10 dataset. The proposed ISANet shows a significant improvement in helicopter (HC) prediction, with its CAM score nearly four times higher than that of other methods. This result validates the effectiveness of the shape branch in refining shape features and constraining geometric structures, making ISANet highly competitive in accurate shape localization. In terms of WSSS performance, it is noteworthy that two methods specifically designed for RS imagery, SEPC and CISM, rank among the top in mIoU compared to methods based on natural images. This indicates a distribution shift between the RS image dataset and the natural image dataset, with methods tailored for RS imagery demonstrating superior performance. Meanwhile, ISANet outperforms other RS image methods by 3.3%–5.2% in mIoU for the segmentation results. This improvement is attributed to the fact that SEPC and CISM do not consider the shape of the objects, an inherent property in RS images, whereas ISANet disentangles shape features from hybrid features for independent learning, thereby enhancing the completeness of activated regions. In this way, the generated object masks are delineated with clearer contours, which is particularly beneficial for handling multiple objects in RS images.

The qualitative comparison results are presented in Fig. 7. The visual results demonstrate that other methods frequently suffer from confusion in segmenting adjacent regions. A representative example can be found in the second row of Fig. 7, where multiple storage tanks are erroneously enclosed by a single boundary, forming an unnatural rectangle. This issue arises from the neglect of shape consistency, as the

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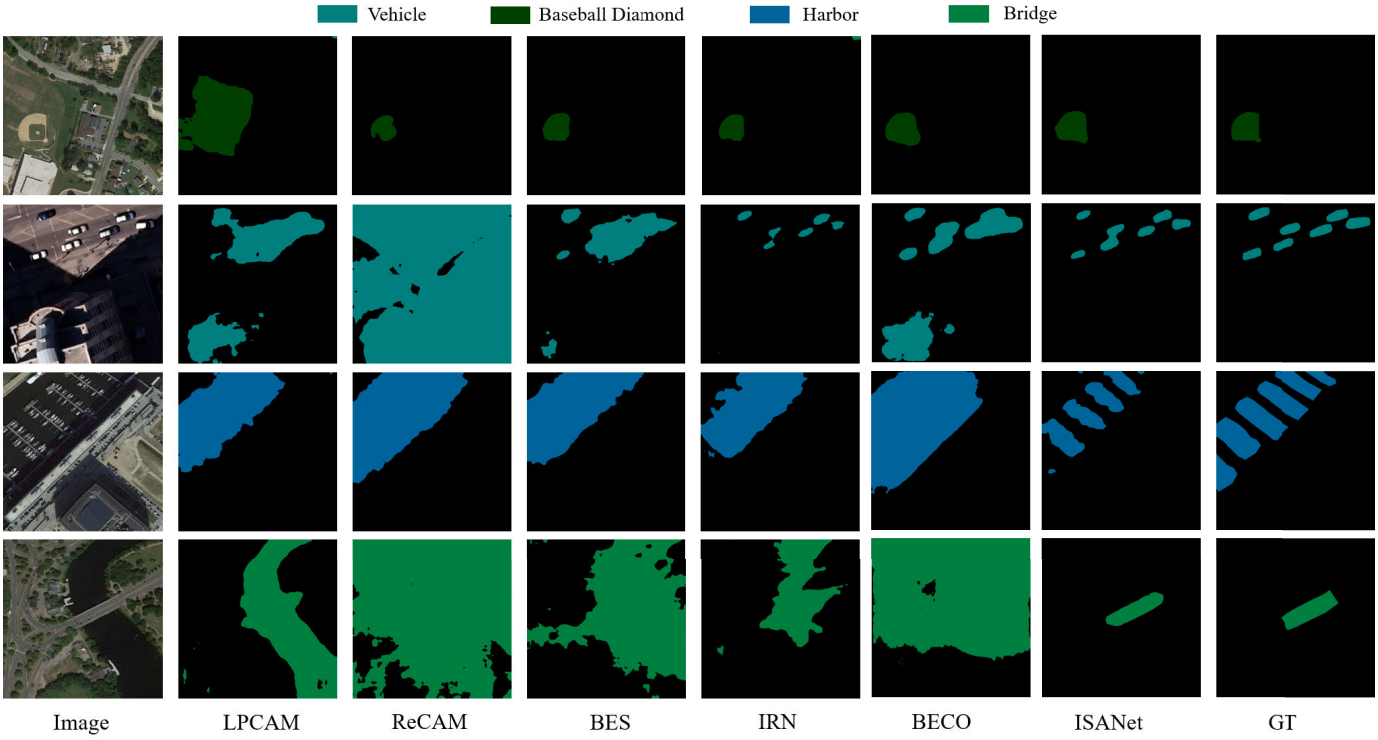


Fig. 6. Examples of semantic segmentation results on the NWPU VHR-10 dataset. We show the results of various sizes of RS objects, including vehicle, harbor, baseball diamond, and ground track field.

TABLE VI
QUANTITATIVE COMPARISON OF OUR ISANET AND OTHER WSSS METHODS ON THE iSAID DATASET

Method	BG	SH	ST	BD	TC	BC	GTF	BR	LV	SV	HC	SP	RA	SBF	PL	HA	OA	mIoU
Performance of CAM																		
RIB [47]	69.9	14.3	32.3	32.2	54.0	35.0	28.4	1.7	11.5	6.0	5.3	3.3	14.3	25.6	21.9	9.6	-	22.8
LPCAM [44]	81.2	19.5	34.4	38.8	39.1	45.9	19.2	2.9	**33.1**	11.2	5.4	6.0	12.3	9.0	24.2	15.4	-	24.8
BES [32]	78.7	11.8	37.3	41.9	57.8	53.3	30.4	1.8	18.3	9.1	6.5	6.1	16.2	**31.7**	23.3	10.1	-	27.1
ReCAM [48]	79.0	17.4	35.8	44.4	62.9	50.5	30.5	1.4	19.2	9.6	5.4	4.6	18.0	28.6	25.5	12.6	-	27.8
IRN [31]	85.3	15.9	40.1	42.7	**66.0**	**54.8**	29.7	2.6	25.5	11.4	5.5	8.0	12.1	23.7	30.1	11.7	-	29.1
ISANet (ours)	**88.3**	**21.6**	**41.7**	**48.5**	62.5	49.4	**32.8**	**4.9**	24.9	**14.4**	**22.0**	**22.7**	**18.2**	25.9	**31.2**	**18.0**	-	**32.9**
Performance of Segmentation																		
SCCAM [49]	76.2	20.0	30.0	37.0	54.3	23.1	28.6	5.7	30.1	11.1	11.9	17.3	16.0	36.1	24.8	10.3	77.3	26.6
BES [32]	83.3	4.1	42.8	56.0	43.5	47.7	12.5	0.7	29.5	11.2	8.5	18.4	14.2	22.0	42.6	11.6	83.5	28.0
LPCAM [44]	84.8	**26.7**	24.7	50.3	36.2	46.2	11.5	0.6	**45.7**	13.9	8.2	12.6	16.5	19.7	40.7	17.2	85.1	28.5
ReCAM [48]	83.5	23.9	38.7	58.6	49.4	45.0	8.8	0.6	28.8	13.5	2.8	13.9	11.7	36.0	43.3	16.3	83.7	29.7
RIB [47]	86.7	16.5	43.6	57.1	60.0	50.0	11.0	0.5	26.0	13.8	11.3	10.4	7.0	20.4	46.9	16.4	86.8	29.8
IRN [31]	87.7	11.7	46.5	62.2	53.0	53.2	11.2	0.6	38.7	9.4	7.9	21.2	10.7	22.8	44.6	15.5	87.8	31.0
CISM [17]	83.1	19.5	44.1	34.4	52.6	48.3	35.5	8.5	31.9	11.8	14.3	17.6	29.2	44.6	30.5	10.8	-	32.3
BECO [28]	89.6	12.1	48.7	61.7	51.4	**54.0**	**38.8**	0.5	43.0	13.7	6.4	7.8	7.3	**47.8**	42.4	20.2	89.6	34.1
SEPC [18]	86.1	17.1	49.1	36.7	56.7	53.3	38.3	**8.7**	30.2	11.5	15.9	19.0	**34.6**	46.7	31.7	11.6	86.1	34.2
ISANet(ours)	**91.1**	13.5	**51.0**	**66.4**	**62.3**	49.7	18.7	1.0	40.3	**14.3**	**34.2**	**40.7**	13.7	24.9	**51.0**	**27.5**	**91.0**	**37.5**

rectangle composed of several storage tanks contradicts the inherent circular shape of a storage tank. Other methods are unable to recognize and correct this inconsistency, negatively impacting the accuracy and reliability of the segmentation results. In contrast, the object segmentation masks predicted by our method exhibit shape consistency, achieving the intended purpose of our network design and aligning with the inherent shape patterns of RS image targets. As shown in the second row of Fig. 7, our segmentation results for each storage tank display a circular contour, which underscores the method's accuracy and reliability in capturing shape features.

E. Results on ISPRS Vaihingen Dataset

To compare the effectiveness of the method in different scenarios, we conducted additional experiments on the ISPRS Vaihingen dataset, a benchmark for semantic segmentation in real-world urban scenarios with complex backgrounds. Unlike the NWPU VHR-10 dataset and the iSAID dataset, which primarily focus on isolated object detection, the ISPRS Vaihingen dataset captures densely built-up urban environments characterized by intricate layouts of buildings, roads, vegetation, and heterogeneous ground objects. We further evaluated the performance of the WSSS of ISANet on this dataset.

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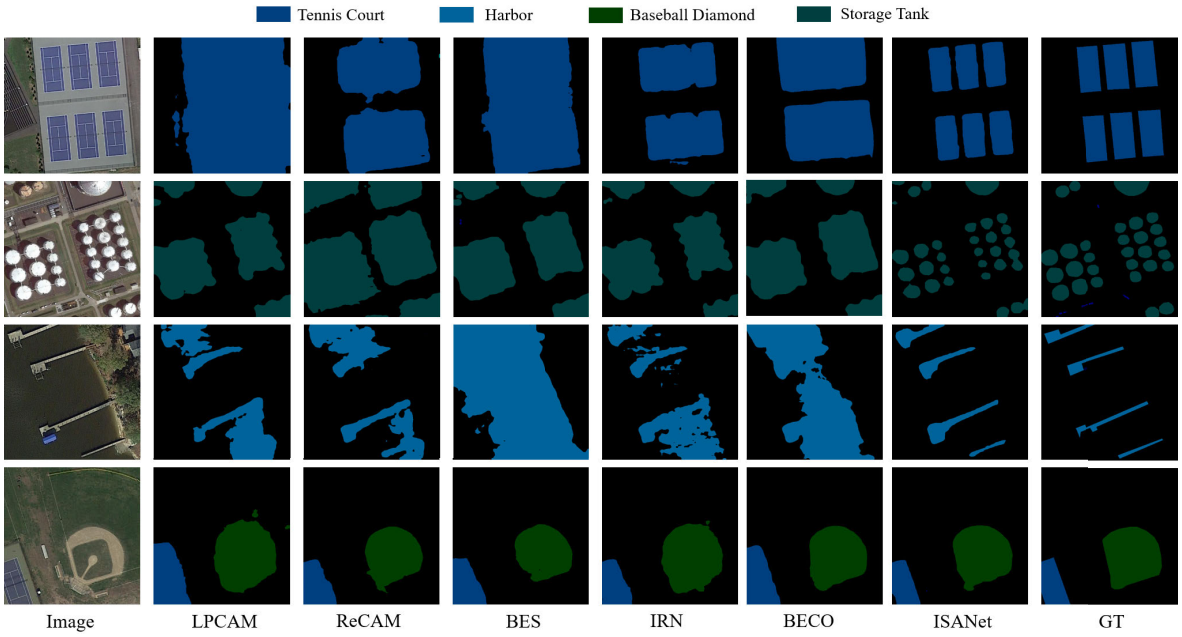


Fig. 7. Examples of semantic segmentation results on the iSAID dataset. We show the results of differently sized RS objects, including tennis court, harbor, baseball diamond, and storage tank.

TABLE VII
QUANTITATIVE COMPARISON OF OUR ISANET AND OTHER WSSS METHODS ON THE ISPRS VAIHINGEN DATASET

models	Impervious Surfaces	Building	Low Vegetation	Trees	Cars	mIoU	OA
Affinity Net [30]	21.45	36.3	28.42	24.91	4.55	23.13	41.06
IRN [31]	33.44	43.79	32.55	45.19	6.83	32.36	53.4
SEAM [50]	35.89	43.71	22.79	51.06	12.12	34.97	53.85
SCCAM [49]	44.03	49.8	38.26	49.13	16.92	41.38	54.34
SEPC [18]	53.86	53.43	44.4	55.79	**19.66**	45.43	65.67
Ours	**68.41**	**74.42**	**54.45**	**59.54**	13.16	**54.00**	**77.64**

The quantitative comparison results on the ISPRS Vaihingen dataset are summarized in Table VII. Despite the unique characteristics of the ISPRS Vaihingen dataset—including densely packed buildings, overlapping vegetation, and intricate impervious surfaces that reflect real-world urban complexity—we observe that the overall performance trend remains consistent with the NWPU VHR-10 dataset.

Our proposed method attains a 54.0% in mIoU and 77.64% in OA, surpassing SEPC by 8.57% in mIoU, with remarkable gains in critical urban classes: +14.55% for Impervious Surfaces (68.41% versus 53.86%) and +21.0% for Buildings (74.42% versus 53.43%). This improvement stems from the geometric priors inherent to urban objects (e.g., regular building contours, linear road structures), which are overlooked by existing methods. By disentangling shape-aware features from texture-dominated representations, our approach generates sharper object boundaries, particularly effective for resolving ambiguities in cluttered scenes (e.g., distinguishing low vegetation from impervious surfaces).

F. Discussion and Analysis

In this section, we discuss the contributions of each components of the proposed ISANet by ablation studies. We also present a sensitivity analysis on the impact of hyperparameters.

TABLE VIII
ABLATION STUDY ON NWPU VHR10 DATASET. *Seg.*: SEMANTIC SEGMENTATION RESULT

No.	Baseline	Shape branch	SAM	CAM mIoU	*Seg.* mIoU
1	✓			40.90	46.02
2	✓	✓		47.30	52.87
3	✓	✓	✓	48.79	53.95

1) Ablation Study: We present extensive ablation studies to validate the individual contribution of different components within ISANet. All experiments are conducted on the NWPU VHR10 dataset. Our baseline model is built with ResNet50, and IRN [31] is adopted to generate the CAM. DeepLabV2 is used as the segmentation network.

In Table VIII, we conducted a comprehensive comparison of the CAMs and segmentation performance across three models: the baseline, our shape branch, and the complete ISANet. Compared to the baseline, the overall performance of our network has been greatly improved. The shape branch significantly improved the mIoU scores of CAM and segmentation results by 3.84% and 6.85%, respectively. The validity of the shape branch supports our new idea of classifying based on shape features. On this basis, to further enhance the

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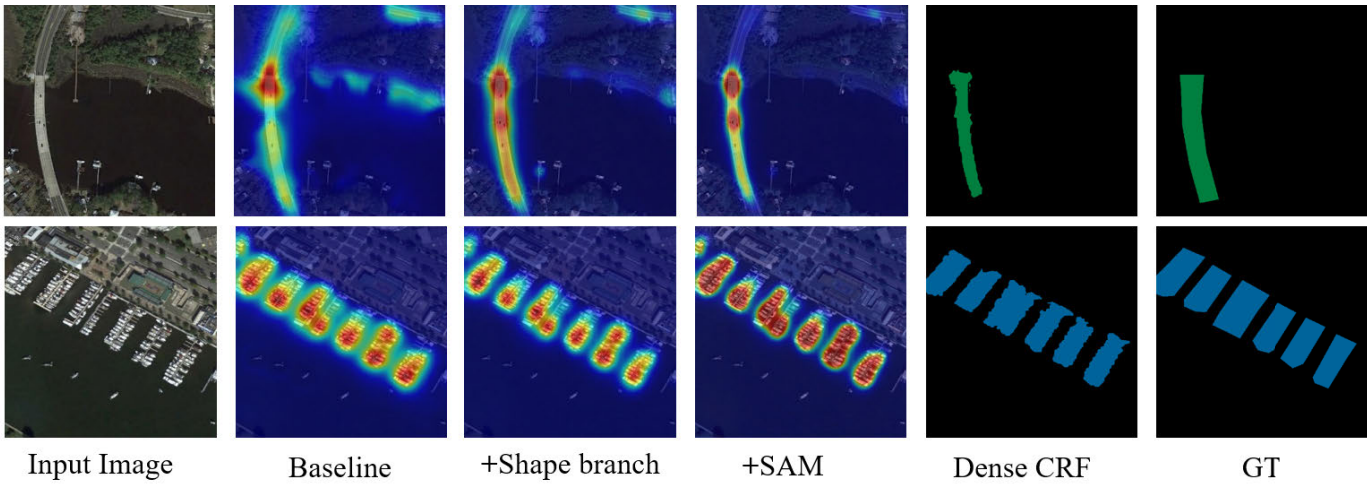


Fig. 8. Qualitative results of the ablation study.

TABLE IX
PERFORMANCE COMPARISON OF POOLING LAYER TYPES ON THE NWPU VHR-10 DATASET

Pooling Type	CAM mIoU	*Seg.* mIoU
Global max pooling	44.28	49.36
Global average pooling	47.30	52.87
Weighted pooling	47.62	53.09
SAM	**48.79**	**53.95**

localization effectiveness of CAM, we add the SAM, with further improvements of 0.56% and 1.08% in mIoU for CAM and segmentation results. This indicates that the geometric shape clues extracted by ISANet and effectively refine the CAM quality, thereby enhancing the accuracy of semantic segmentation.

To validate the effects of the SAM, we conduct fine-grained ablation experiments comparing four pooling strategies on the NWPU VHR-10 dataset. As shown in Table IX, SAM achieves the highest performance with 48.79% CAM mIoU and 53.95% Seg. mIoU, significantly outperforming standard pooling methods and weighted pooling. Global max pooling exhibits the weakest results, as its overemphasis on local maxima. GAP and weighted pooling further enhance segmentation accuracy through context aggregation. Furthermore, SAM incorporates a shape-aware self-attention mechanism to refine pixel-level geometric alignment, achieving the best results.

In order to demonstrate the effects of our modules more intuitively, we add shape branch and SAM module to the baseline in turn. The visualization results of CAMs with different stages are shown in Fig. 8. When an RS image is input, the baseline model without any modules struggles to accurately activate targets within the object shape range. Following the addition of the shape branch, all targets are more accurately located, and the activation regions became more similar to the actual target shapes. Finally, the SAM module eliminated the highlighting of background areas without any targets, preventing interference from background noise in the generation of pseudo-labels. Thus, with the constraint provided by shape priors, the ISANet-based CAMs and pseudo-labels possess a more shape-fit object delineation, and the boundaries between multiple targets are more finely separated.

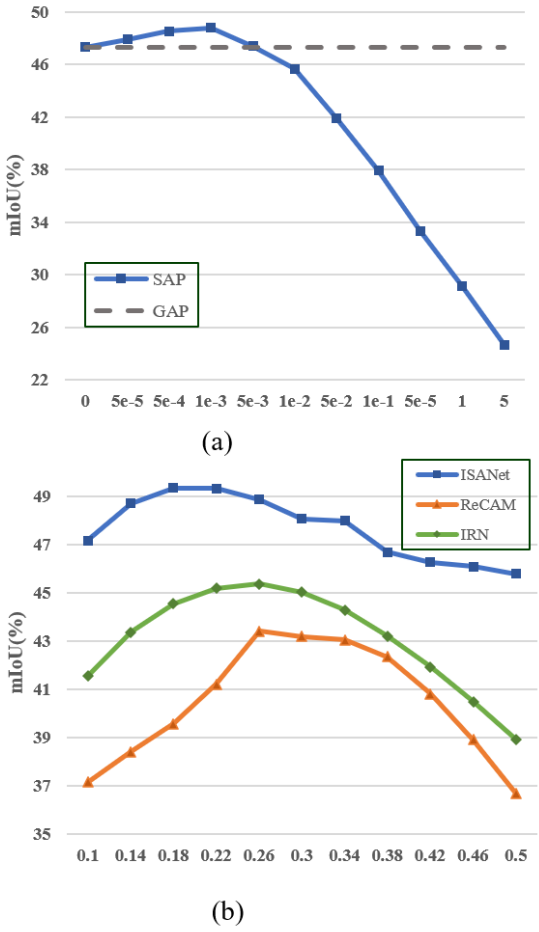


Fig. 9. Sensitivity analysis for hyperparameters on NWPU VHR-10 dataset. (a) Parameter k for adjusting the intensity of SoftMax function. (b) Threshold used to generate 0-1 mask from CAM.

2) Sensitivity Analysis for Hyperparameters: In this section, we present a hyperparameter sensitivity analysis on three

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TABLE X
IMPACT OF USING VARIOUS RESOLUTION FOR THE SHAPE FEATURES TESTED ON NWPU VHR-10 DATASET

Resolution	256	128	64	32
mIoU(%)	48.83	48.79	47.94	47.28

parameters: the parameter k for adjusting the intensity of SoftMax function in (5), the threshold used to generate 0-1 mask from CAM (a general hyperparameter in CAM-based methods), and the resolution of the shape features.

Fig. 9(a) shows that, compared to GAP-based CAM, the proposed attention-pooling CAM achieves a slight improvement when the attention intensity parameter k is set to a small value. However, further increases in k result in a rapid decline in performance. This decline occurs because when k becomes sufficiently large, SAM behaves similar to max pooling, which disregards the contributions of other pixels. As shown in Fig. 9(b), ISANet exhibits much gentler slope around its respective optimal points compared to other methods. This observation indicates that ISANet is less sensitive to variations in the threshold parameter, demonstrating robustness and stability across different settings. As shown in Table X, our method demonstrates robustness in terms of the resolution selection. Considering the trade-off between performance and efficiency, we chose a resolution of 128 for the shape features.

V. CONCLUSION

In this article, we propose an ISANet to tackle the issue of unconstrained region response learning in WSSS of RS images. We revealed the existence of strong correlation between object shapes and their semantics in RS imagery and validated the importance of shape features in inferring the semantics of RS objects. Extensive experiments conducted on two challenging benchmark datasets, i.e., NWPU VHR-10 and iSAID, show that the proposed ISANet can effectively distill the inherent, semantic-meaningful shape patterns in RS images and thereby generate more complete CAMs with clear boundaries. With these high-quality CAMs, the proposed ISANet significantly enhances the performance of WSSS, achieving 7.9% and 3.3% mIoU improvement compared to the state-of-the-art method on the NWPU VHR-10 and iSAID datasets. Comprehensive ablation studies further verify that the effectiveness of the shape branch and SAM module, which effectively disentangle the shape features from the mixed ones during training. In the future work, we would like to explore the shape consistency and the unmixing shape learning mechanism in other RS tasks.

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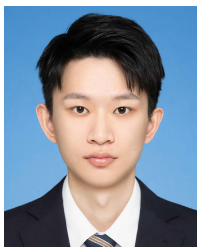
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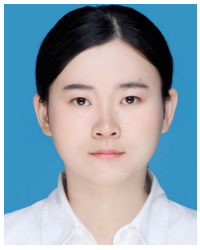
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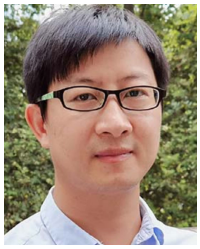
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